



VALUES • EDUCATION • SERVICE

Summer 2026

Course Information

Number: ONL-NURS-710-A0

Title: Biostatistics

Delivery Method: Online

Section: A0

Time and Place: The online weeks each begin Monday @ 8am EST and close Sunday at midnight EST

Credit Hours: 3 Credit Hours

Instructor Information

Name: Jennifer Stewart-Glenn, PhD, MSN, FNP-BC

Email: J.stewart-glenn@lmunet.edu

Phone: (c) 276-690-6746

Office Location: Appointments available by phone or Zoom

Office Hours: As posted or please email to schedule a time

Email Response Time: 24-48 hours during business days

Course and Program Specific Information

Course Description

This course provides an overview of the most commonly used biostatistical methods in the context of nursing science and health care related research. The major topics to be covered include types of data, descriptive statistics and plots, theoretical distributions, probability, estimation, hypothesis testing, and one-way analysis of variance. An additional focus is the use of biostatistics to identify, evaluate, and implement data that contribute to meaningful evidence in populations based research, evidence-based advanced nursing practice, and optimal health outcomes.

Course Objectives

Upon completion of this course, the student will be able to:

1. Examine types of data and principles of selected descriptive and inferential bio-statistical analyses and application to nursing science and health care research, including meaningful translation into evidence-based advanced nursing practice.
2. Correlate appropriate use of biostatistics contributing to meaningful evidence in populations based research, advanced nursing practice, and optimal health outcomes.
3. Appraise data analysis and interpretation of statistical results in extant nursing and health care literature.
4. Translate nursing science knowledge from biostatistics in the context of clinical practice guidelines that contribute to analysis of practice relevant data, and predict health care outcomes.
5. Analyze biostatistical measures that promote validity, reliability, and generalizability of measured variables and or psychometric properties in hypothesis testing.

Texts/Materials for the Course

REQUIRED:

American Psychological Association (APA). (2019). *Publication manual of the American Psychological Association* (7thed.). Washington, DC: American Psychological Association.

Statistics Intellectus© software access (purchasing information posted in Canvas)

*Select journal articles, other readings and materials as posted in Canvas Modules

Course Requirements, Learning Outcomes, and Evaluation Methods

1. **Course Week:** The electronic week begins on Monday at 8:00am Eastern Standard Time (EST) and ends on Sunday at midnight Eastern Standard Time.
2. **Canvas:** Canvas is the online learning platform used for this course. Students must have proficient computer skills necessary to participate in the course. The online Module week begins on Monday at 8:00 am Eastern Standard Time (EST) and ends on Sunday at midnight (EST).
3. **Attendance and Communication Requirements:** Students are required to log on to the course Canvas site on a regular basis and participate in the discussion board and complete any assignments, quizzes, and exams by the due dates. Course announcements and LMU email should be checked at least daily, and students should respond to faculty email communication within 24 hours.

In the event of extenuating circumstances, students should notify course faculty prior to class to make arrangements for make-up work. Failure to do so may affect the course grade. Online class attendance will be assessed by completion of all online course assignments by the published due date and time. Students are responsible for retrieving course messages and announcements from both their LMU email account and through course announcements in Canvas.

4. **Student Health Insurance and Clinical Health Requirements:** Personal Health insurance and maintaining clinical health requirements is required for all students enrolled in a NURS course. If proof of health insurance or clinical health requirements are not provided as required, the student will be withdrawn from all NURS courses.

Information Literacy/Technological Resources

Technology is an essential and integral part of an online course. The student must have a laptop computer no more than 2- 3 years old and a stable connection to a high-speed internet such as a cable modem or digital subscriber line (DSL) on a regular basis. In general, Windows based computers are best as support is currently limited on Apple products. The student laptop should have Windows 7 or above operating system. The student will need access to MS Office 2013 including MS Word, PowerPoint, and an Excel software programs. The student will need the capability of audio access for some lectures and materials. A webcam and microphone are also required. More than one browser should be installed on the student laptop, recommended browsers are: Internet Explorer, Google Chrome, and Firefox (Certain aspects of Canvas and the library databases work best in certain browsers). Browsers should be kept up to date and tested to ensure the student has the necessary browser capabilities. The browser test is located on the LMU tab of Canvas. The general recommendation for Canvas is use of Firefox or Google Chrome with a hard-wired connection if at all possible. A hard wire connection is particularly important for online testing and synchronous course materials. The primary and preferred method for verification of student identity, for distance and online education purposes, is the use of a secure login and passcode. Other options include proctored examinations and the use of approved technologies. The Respondus Lockdown Browser, which is available for download on Canvas, should be installed. Additional help and information on informational literacy and technology resources and support is available at the helpdesk at 423-869-7411.

Bring your own device (BYOD) is a policy that allows students to bring personally owned mobile devices onto the campus to access privileged LMU resources such as email, file servers and databases as well as their personal applications and data. LMU IS department highly recommends that each student bring a personal computing device to campus with them that includes:

- High-speed (broadband) Internet connection with a minimum bandwidth of 5 Mbps upload and download speed,
- Microsoft Office (Available by contacting the Information Services Helpdesk at 423.869.7411),
- Microphone and speakers, a webcam is recommended, and
- Windows (recommended) or Mac computer with at least 8 GB of random-access memory (RAM) and 259 GB of free hard-drive space,
- Web browsers: Google Chrome and Mozilla Firefox (Install both browsers).
- In order to maintain full compliance with the LMU Network Policy, the LMU IS Department utilizes a Network Access Control Appliance (NAC). The NAC requires each user to register each device and bring that device into compliance before being granted network access.
- **It is not recommended to complete your coursework entirely on iPads, iPhones, Android phones, Android tablets, or Chromebooks. A laptop or desktop computer is best for completing classwork in Canvas.**

Please refer to <https://www.lmunet.edu/information-services/i-need-help-with/> for more information related to information literacy and technological resources.

STUDENT EMAIL POLICY

Every student is issued an LMU email account. Electronic mail (email) is an official mechanism for administrators, faculty, staff and students to communicate with each other. The University expects that email communications will be received and read in a timely manner. Students are expected to check email on a frequent and regular basis in order to stay current with University related communications, recognizing that certain communications may be time-critical. If a student receives an official email from a University faculty member, administrator, or staff member and does not read that email,

any subsequent repercussions cannot be excused by “unread email messages.” Inappropriate emails are prohibited. Anyone receiving such an email should immediately contact the University Helpdesk.

The following Policies can be found in the *DNP Handbook 2025-26* under Code of Ethics Violations

Plagiarism Policy: Plagiarism occurs when an author uses someone else’s original work without acknowledgement. Whether intentional or unintentional, plagiarism is a violation of Lincoln Memorial University’s Academic Integrity policy. As scholars, students have an absolute obligation to provide accurate information about the origin of ideas, information, graphics, or images presented in all formats. The obligation acknowledgement applies to all scholarly products including, but not limited to, papers, presentations, slides, and posters.

Use of the work of another without acknowledgement is the most serious form of plagiarism. Errors in proper acknowledgement can range from isolated typographical, formatting or style errors, to more serious issues with inadequate paraphrasing or patterns of problems within an assignment. The plagiarism policy of the Caylor School of Nursing allows faculty members at the classroom level to distinguish among levels of seriousness when deciding on an appropriate penalty. Penalties can range from loss of points related to formatting, to a penalty grade reduction for the assignment, to assigning a failing grade for either the assignment or the course.

The citations within a scholarly work are the means by which credit is given to the original source. Citations also allow an interested reader to seek out the original source of material. In the process of writing a paper, students (both undergraduate and graduate) are expected to synthesize information and ideas from multiple sources to create something new. The process most often requires paraphrasing rather than quoting from an original source. Direct quotations should be reserved for those cases where language of the original sources is unique or particularly well written. While not plagiarism, a paper that consists of a series of quotations does not meet the standard for college-level work.

When paraphrasing, it is important to remember that it is inappropriate to change a few words while retaining the same sentence structure or paragraph formation as an original source. Authors express ideas by choosing words, building sentences, and constructing paragraphs to explain and support ideas. Even if the source is cited, it would be a form of plagiarism to use the structure of an author’s work as if it were one’s own writing. One of the best ways to avoid struggling with Plagiarism is to use (and cite) multiple sources and put one’s thoughts into one’s own words. The student should avoid reliance on a lone source, as it is much easier to decide where to place citations within the body of paper with the use of multiple sources.

Recommended Resources on Plagiarism: There are multiple resources available to illustrate the appropriate use of American Psychological Association (APA) citation and reference list requirements.

If the student has questions or concerns about how to incorporate a particular source into one’s own work, the student should initiate a dialogue with an instructor regarding the questions or concerns. The student should use faculty as a resource to answer a question or complete a significant amount of the assignment well in advance of the due

date.

When a student receives feedback on a graded assignment and is unclear about how to avoid a future problem, he/she should make an appointment to discuss the issue with the faculty member or consult with the writing center.

Plagiarism will be addressed in the following manner:

1. Faculty members at the classroom level will distinguish among levels of seriousness when deciding upon appropriate penalties for plagiarism. Guidance and expectation regarding penalties for plagiarism will be addressed in class and in the course syllabus.
2. Students should expect to receive a failing grade on any assignment which contains significant amounts of plagiarized content.
3. In cases where an entire paper is plagiarized or where the student represents another student's paper as his or her own, can result in the instructor assigning an "F" for the course.
4. Students are not permitted to withdraw from a course in which they have been charged with plagiarism and/or cheating for any reason.

Use of Generative Artificial Intelligence (GenAI)

The use of Generative Artificial Intelligence (GenAI) in any work submitted as assignments in class, labs, clinical work, and/or for other course requirements is prohibited. All work is to be entirely your own unless otherwise permitted by the instructor. Any unauthorized use of GenAI will be considered a violation of the Caylor School of Nursing Code of Ethics.

LMU Caylor School of Nursing Online Testing Policy

Rules Regarding Online Tests and Examinations: Quizzes and exams will be available on Canvas. Online quizzes, exams, and final exams will require Respondus LockDown Browser. Students are required to complete these assignments in a time-sensitive fashion. Students are required to complete these assignments within the assigned time frame. All exam dates will be announced per course. The dates are included in the course schedule.

Violation of ANY of the following policies will result in a zero (0) for that exam.

LMU Caylor School of Nursing Online Testing Policy

A. All students are expected to take exams as scheduled.

1. Technical requirements:

- Technology is an essential and integral part of online testing. The student must have a laptop computer no more than 2-3 years old on a stable connection to a high-speed internet such as a cable modem or digital subscriber line (DSL) during online examinations. A webcam and microphone are also required. Browsers should be kept up to date and tested to ensure the student has the necessary browser capabilities for exam purposes. All exams will be given via online LockDown Browser unless otherwise specified in the course syllabus.

2. If unable to complete or take the exam: Students must notify the faculty by phone or email (see Faculty Contact Information) prior to the scheduled exam if they cannot take an exam (for reasons of sickness, etc.). Students are given faculty contact information in each NURS course syllabi and are expected to always have it available.

- If for any reason a student is unable to leave a message for the faculty member via the contact information provided, it is the student's responsibility to contact the Nursing Office in Harrogate (1- 800-325-0900, ext. 6324) or the appropriate site location and talk to an administrative assistant or leave a message

on the voice mail. Please remember to state you are unable to take the exam and be specific as to the course, the faculty's name, and the site you attend. **Any student that does not notify the appropriate faculty may receive a zero for the exam.**

- Faculty will determine the date, time, and method of any alternative make-up exam. A pattern of missing exams in a specific course or throughout the program will not be tolerated. A counseling record will be written with each missed exam, and the specific consequences of another missed exam will be provided. If the student does not make up the exam on the scheduled date and time, the student will receive a zero on the exam.

3. During the exam: At LMU, the primary and preferred method of verification of a student's identity for online education is the use of your username and password and student ID.

- Please be sure you have the Respondus LockDown Browser installed (Instructions for download on the Canvas site-must be LMU version) on your device, a webcam, and LMU ID available prior to the exam start time.
- The exam will be timed. Once the ID verification the exam timer will begin. You will have only one opportunity to enter and complete the exam.
- If you should run into technical difficulties during the exam, speak to the camera and say that you are using your phone to contact the instructor or the help desk.
- Videos will be reviewed and any suspicion of cheating will be investigated.
- Student questions related to exam content will not be answered by faculty during the exam.
 - Students are not to talk or communicate with another person during the exam. If talking/communication occurs, the students involved will be subject to the University's cheating policy.
 - The student is prohibited from sharing any information with any other individual or student in written, verbal, electronic, photographic, or other format at any time during the exam period or the semester. Sharing of any quiz or exam information will be considered a form of academic dishonesty/ cheating and will follow the disciplinary proceedings described in the Graduate/ Undergraduate Student Handbook. The instructor reserves the right to investigate any potential sharing of information in the above format anytime during the semester.

4. After the exam: An item analysis of all questions is completed before the exam grades are determined and released.

- You may review your exam according to the specific course syllabus.
- Challenges to exam questions will be accepted according to course syllabus.
- Challenges should be sent in an email format and should specify the rationale as to why the chosen answer should also be considered as a "correct answer." Appropriate references from course materials and documentation should be provided with rationale. Challenges concerning the appropriateness or relevance of the question are not within the purview of the student and will not be considered.
- Nursing faculty will have one week to review and score exams. Individual student grades will be available and posted on Canvas within one week after the exam has been given.
- For the last exam of the semester (final unit exam or final comprehensive exam), students must contact the instructor within 24 hours for clarification of any exam related issue.
- Faculty reserves the right to correct any clerical error. This includes both increases and decreases to adjusted exam grades.

5. Please refer to the DNP Handbook and other sections of the syllabus for course specific exam policies that may be in addition to the above.

Failure to comply with any exam policy may result in a zero on the exam and possible dismissal from the program.

Violation of ANY of the following policies will result in a zero (0) for that exam.

Student Health Insurance

All graduate nursing students must have medical insurance while enrolled in any of the Caylor School of

Nursing programs. Students must submit adequate documentation demonstrating coverage of health insurance. Students are required to notify immediately the Graduate Nursing office of any change in health insurance provider or coverage. Students will be charged a fee by the finance department until proof of active insurance is provided.

Graduate Grading Scale

The LMU grading system is based on a four-point scale. A grade of "B" or better is required to be successful in this course and to receive graduate credit at LMU.

The grading scale is as follows:

A	93-100	4.00 quality points
B+	89-92	3.33 quality points
B	83-88	3.00 quality points
C+	79-82	2.33 quality points
C	73-78	2.00 quality points
D+	69-72	1.33 quality points
D	63-68	1.00 quality points
F	Below 63	0.00 quality points

* Partial points will be rounded to the nearest full point; for example, 88.49=88 leads to a grade of B; and 88.50=89 leads to a grade of B+. Partial points are rounded for purposes of the final grade of the course.

Incomplete Policy

A grade of Incomplete ("I") may be given in circumstances where the student has successfully completed the majority of the coursework and the instructor determines that exceptional circumstances warrant extending the time for the student to complete the work. In order to receive an "I", the student and course instructor must negotiate a contract containing a timeline specifying the date(s) by which the remaining coursework will be completed (including deficient practice hours) and the written assignments or examinations to be completed. *The form for documenting a contract for an "I" is found in the DNP Handbook.* It is the responsibility of the student to monitor progress towards completion of the contract to remove the Incomplete and to arrange for make-up assignments, exams, labs, or clinical hours.

If a grade change request has not been submitted by the instructor by the required date, the "I" grade automatically becomes an "F" on the student's transcript.

In general, an "I" grade in any sequential nursing course must be removed by the end of the first six weeks of the next semester in order for the student to continue enrollment in the next nursing course in the sequence. In exceptional cases where serious illness or temporary disability prevent completion of the clinical component of a course but the student is able to successfully continue with the didactic component, the student may petition the Dean to negotiate an alternative plan to complete all requirements for removal of the "I." Before granting such a waiver, the Dean will require a statement from the treating physician specifying that there is a reasonable expectation that the student will be able to complete the clinical component and remove the "I" within the proposed time line. An alternative would be to request a medical withdrawal.

Medical Withdrawal

If at any time during the term a student is unable to perform all of the Student Essential Functions listed in the CSON DNP Student Handbook for a period exceeding two weeks, the appropriate course of action is to medically withdraw from the class. As an alternative to withdrawal, within two weeks of becoming unable to perform the essential student functions, a student must provide documentation from a medical provider that the student is able to perform all of the Student Essential Functions listed in the CSON DNP Student Handbook. Students will need to contact the concentration director if seeking a Medical Withdrawal.

Course Assignments:

1. Students must complete scheduled assignments by the posted due dates, including as indicated under "Discussion Board" and "Methods of Evaluation." Failure to submit assignments, quizzes, exams or participate in the discussion board by the posted due dates will result in a zero grade.
2. The course assignments are listed in the course schedule in this syllabus and in more detail in individual Canvas Modules. There may be additional articles or resources that are posted throughout the semester for additional reading, posted within the Canvas weekly modules.
3. Detailed instructions for any assignments, quizzes, exams and discussions are located in each module.

Methods of Instruction

Methods of instruction may include but are not limited to:

Written Assignments

Zoom

Recorded lectures

Required & Recommended Readings

Video Tutorials

Quizzes and Associated Learning Activities

Analysis exercises using Statistics Intellectus

Information provided for the course by whatever means is intended only for those that are enrolled in the course. The student is prohibited from sharing any information with any other individual or student in written, verbal, electronic, photographic, or other format at any time during the semester. This includes but is not limited to PowerPoints, PDF files, links to PowerPoints, YouTube videos, lecture materials, exam review, discussion materials, etc. The instructor reserves the right to investigate any potential sharing of information in the above format anytime during the semester.

Methods of Evaluation

Knowledge and competence of theory will be evaluated through successful completion of class assignments including the presentation assignment, discussion board posts/responses, and Statistics Intellectus quizzes and modules.

Discussion Board/ Assignments (Completed as Individual)	35%
Statistics Intellectus Assignments/ Quizzes (Completed as Individual)	35%
Instrument Evaluation Presentation (*option to be completed as a group)	30%
Total	100%

All assignments, quizzes, presentations, and activities must be completed to complete the course and not completing any of these on time will result in a zero.

Discussion Board And Other Assignments Discussion assignments are in select modules and participation is required. Any omissions will result in a zero for that discussion. Detailed grading guidelines for the discussion board posts are located in the appendices of this syllabus and the appropriate weekly module in

Canvas. Unless otherwise noted in the instructions, initial discussion posts are due by Wednesday at 11:59 PM EST and response posts are due by Sunday at 11:59 PM EST. Follow the guidelines in Canvas and in the appendices of this syllabus carefully to receive maximum points. The only other assignment in this course is completing the course sign up page with your presentation and group information.

Statistics Intellectus Assignments/Quizzes The course requires access to Statistics Intellectus and contains assignments and quizzes using Statistics Intellectus with associated learning materials. The Intellectus quizzes allow unlimited attempts before the due date. Any attempts after the due date will not be counted. Some of the quizzes may require analysis using Intellectus to answer questions and the learning materials associated with the quiz provides an overview.

Instrument Evaluation Presentation In a presentation format the student will evaluate and appraise an instrument, scale, or tool of their choice based on clinical phenomena or concepts of interest. Details are located in the appendices at the end of this syllabus and materials in Module 2. Students may opt to complete this assignment as a group and topics must be submitted for faculty approval.

No extra credit is given in this course. Passing is considered a weighted average Of 83. All assignments must be completed to pass the course.

Course Outline/Assignments/Units of Instruction or Clinic Schedule

Date	Topics/ Readings/ Materials **Additional topics, readings. or materials may be posted	Tasks and Assignments Due
Module 1 Week of May 11th	Syllabus and course overview Access Statistics Intellectus	Discussion Board: Introductions by May 17 th 11:59 PM EST Purchase Statistics Intellectus and set up account
Module 2 Week of May 18 th	Select Readings posted on measurement, validity, reliability, and psychometric and/or instrument testing. Review of criteria for Instrument Evaluation Presentation	Live ZOOM Session May 19 th at noon EST No assignments Due this week ** Be thinking about if you would like to work in a group and potential presentation concept/instrument you will be asked to submit this for faculty approval and feedback due in Module 3
Module 3 Week of May 25 th <i>*note LMU is closed for Memorial Day May 25th</i>	Fundamentals of Statistics Part 1: Levels of Measurement, Types of Variables, Data Collection and Measurement, Descriptives	Intellectus Lessons/Quizzes due by 11:59 PM May 31 st Assignment: Group Information and Concept/ Selected Instrument due on sign up page by 11:59 PM May 31 st
Module 4 Week of June 1 st	Fundamentals of Statistics Part 2: The Research Process, Hypothesis Testing, Sample Size and Power Analysis,	Intellectus Lessons/Quizzes Due by 11:59 PM June 7 th Discussion Assignment: Research Process

	Probability, Choosing the Correct Statistical Test	initial post due by 11:59 PM Wednesday June 3 rd , 1 response post due by 11:59 PM Sunday June 7 th
Module 5 Week of June 8 th	Illustrating Data	Intellectus Lessons/Quizzes due by 11:59 PM June 14 th Discussion Board Assignment: Initial Post Due by 11:59 PM Wednesday June 10 th , 1 Response Post Due by 11:59 PM Sunday June 14 th
Module 6 Week of June 15 th	Research Design, Z-test, T-test	Intellectus Lessons/Quizzes due by 11:59 PM June 21 st
Module 7 Week of June 22 nd	Bias, Correlation	Intellectus Lessons/Quizzes due by 11:59 PM June 28 th Discussion Assignment: Initial Post Due by 11:59 PM Wednesday June 24 th , 1 Response Post Due by 11: 59 PM Sunday June 28 th
Module 8 Week of June 29 th <i>*Independence Day Holiday Week</i>		No assignments Due
Module 9 Week of July 6 th	ANOVAs, MANCOVAs	Intellectus Lessons/Quizzes Due by 11:59 PM July 12 th
Module 10 Week of July 13 th	Non-parametric Tests Regression	Intellectus Lessons/Quizzes Due by 11:59 PM July 19 th
Module 11 July 20 th	Instrument Evaluation Presentations	Instrument Evaluation Presentations due to be Posted in Discussion Board by July 20 th 8:00 AM and Submit a copy of your presentation in the submission area by July 20 th 8 am EST. Discussion Board Assignment (presentation counts as an initial post, 1 response posts is due by next week on July 29 th 11:59 PM EST to give you time to review each of the presentations.
Module 12 Week of July 27 th	Finals Week Posted Supplemental Materials on selecting statistical tests	Response post from Module 11 Presentations Due by July 29 th 11:59 PM EST.

Important Dates in the Academic Calendar Summer 2026

Event	Date(s)
Summer Term	May 11-July 31
Memorial Day (no classes)	May 25
Independence Day (no classes)	July 3

Information for All Courses and Programs

University Services

Academic Support Services

The [Tagge Center for Academic Success](#) offers tutoring options, study skills sites, student support services, writing support, and the Cornerstone Program.

Counseling and Accessibility

Counselors are available to help students with personal, career, and academic concerns that affect academic success and quality of life. The Executive Director for Counseling and Accessibility, Jason Kishpaugh, can be contacted at jason.kishpaugh@lmunet.edu and/or 423.869.6277 (800-325-0900 ext. 6277).

University Policies

Undergraduate Attendance

To maximize the learning experience at Lincoln Memorial University, students are expected to attend all classes. All absences, excused or unexcused, must be brought to the attention of the instructor as soon as possible. All missed coursework, tests/assignments/information/clinical activities, are the student's responsibility. LMU's institutional Attendance Policy (including a list of excused absences) is published in the [Academic Catalog](#).

Unofficial Withdrawals

Any student who ceases attending classes before the end of the semester, or summer term, without completing an official withdrawal from the University, automatically receives the grade of "F" for such course(s), so noted on the student's academic transcript, and may be administratively withdrawn. LMU's institutional policy on Unofficial Withdrawals is published in the [Academic Catalog](#).

Administrative Withdrawals

Students who have not attended courses by the ninth class meeting of the semester (or equivalent for summer terms) will be reported to the Registrar's Office, Financial Aid, and the Tagge Center for Academic Success and may be administratively withdrawn with a WD recorded on the transcript for each course. Students who cease attending classes prior to the end of the semester, mini-term, or summer term without completing the official withdrawal from the University may also be administratively withdrawn, with an F recorded on the transcript for each course. (See "Unofficial Withdrawal.")

Distance Education Policies (located on LMU Policies [webpage](#))

The following policies are in place for students in distance education (online) courses or programs.

[Verification of Identity](#)

[Protection of Student Privacy](#)

[Notification of Additional Charges Associated with Student Identity Verification](#)

Students With Disabilities Policy

LMU is committed to providing reasonable accommodations to assist students with disabilities in reaching their academic potential. If you have a disability that may impact your performance, attendance, or grades in this course, please contact Dr. Jason Davis, Director for Accessible Education Services, to discuss your specific needs at jason.davis@lmunet.edu and/or 423.869.6587 (800-325-0900 ext. 6587). Additional information regarding accommodations is available on the [Accessible Education Services webpage](#).

Please Note: The Department of Accessible Education Services strives for prompt communication. If you have not heard back from us in 5 business days from the date of this email, please email our department again.

Discrimination and Academic Integrity

Policies can be found in the [Railsplitter Community Standards Guide](#) and [Academic Catalog](#). Note: some graduate/professional programs have program-specific Academic Integrity policies, which students in those programs are required to follow.

Harassment, Discrimination, and Sexual Misconduct

Lincoln Memorial University prohibits discrimination on the basis of race, color, ethnicity, religion, sex, national origin, age, ancestry, disability, veteran status, sexual orientation, marital status, parental status, gender, gender identity, gender expression, and genetic information in all University programs and activities. The [LMU policies on harassment, discrimination, and sexual misconduct](#) are located in the [Railsplitter Community Standards Guide](#) and through the [Office of Institutional Compliance](#).

All LMU employees, except counselors in the LMU Office of Mental Health Counseling in the course of providing clinical counseling, are Mandatory Reporters for sexual harassment on campus. This means that if you inform any LMU employee of a situation that may involve sexual misconduct, including sexual harassment, sexual assault, stalking, domestic violence, dating violence, or any other form of prohibited gender or sex-based discrimination, the employee is required to report the information to the Title IX Coordinator.

If you have experienced discrimination and would like to make a report to the University or discuss your options for support from the University, contact:

Ms. Rebekah Webb, M.Ed., Title IX Coordinator & Institutional Compliance Officer

Phone: (423) 869-6315 or Emails: rebekah.webb@lmunet.edu or TitleIX@LMUnet.edu

Office: LMU Cumberland Gap Offices #3 (next-door to the Lincoln's Closet) | Duncan School of Law 249 (Knoxville) | Meet by Appointment on any LMU campus or via Zoom.

**If you need pregnancy accommodations, please contact the above Title IX coordinator.*

Hazing

Hazing is strictly prohibited by the University and the State of Tennessee. Additional information on LMU's policy on hazing is published in the [Railsplitter Community Standards Guide](#).

Course Evaluations

In addition to meeting degree requirements specified in the graduate and undergraduate catalogs, all students are required to complete University-administered course evaluations.

Outcomes Assessment Testing

Degree requirements include participating in all outcomes assessment testing (e.g., general education assessment, major field tests, etc.) and activities when requested. Students may be required to complete one or more questionnaires and to take one or more standardized tests to determine general educational achievement as a prerequisite to graduation (see appropriate catalog for additional information). More information is available on [LMU's Assessment webpage](#).

Inclement Weather

In the event of inclement weather, students should check their LMU email during delays/closures to receive information from individual faculty regarding potential assignments and/or other course information.

Instructional Continuity in Case of Temporary Campus Closure Policy

Faculty and students should expect scheduled instruction to continue even if class meetings are canceled due to weather, epidemic illness, or other factors. Students will be required to complete alternate instructional activities online as directed by the course instructor.

Mission Statement

Lincoln Memorial University's Mission Statement is located on the [Heritage and Mission webpage](#).

Student Community Engagement

A cornerstone of the University's mission is service to humanity. For more information visit [LMU's Leadership and Outreach webpage](#) or contact Assistant Dean of Students Sheri Schatz at sheri.schatz@lmunet.edu.

Originality Checking Tools

Students agree that by continued enrollment in this course that all required papers may be subject to submission for textual similarity review to third-party tools for the detection of plagiarism.

The instructor reserves the right to revise, alter, and/or amend this syllabus, as necessary. Students will be notified in writing and/or by email of any such revision, alterations, and/or amendments.

APPENDIX A

Discussion Board Guidelines

Select modules within this course will have discussion board assignments (see course schedule and modules in Canvas). Details for discussion board assignments will be posted in the weekly modules along with other course materials for that week. **The student's initial response to the posted materials is due by Wednesday at 11:59 pm of each week** with an assigned discussion unless otherwise noted in the schedule. Exceptions for this course include 1) Introductions which are not due until Sunday at 11:59 PM in Module 1 and the presentation discussion in which the initial post is the posted presentation and is due by 8:00 am at the start of Module 11 with extended time given for the response posts to allow for presentation reviews (see directions).

In addition to the initial post, the student **must respond to a minimum of one post before the discussion board closes on Sunday of that week at 11:59 pm (unless otherwise noted due date)**. Posts must be substantive and reflect critical thinking and deep reading of the posted materials. It is not acceptable to simply reply "good post" or "I agree." Posts must reflect higher-level thinking as reflected in the table of Bloom's Taxonomy below to receive credit for participating in the discussion. **Initial posts require 150 words or more and each response to a peer requires 100 words or more.**

Do not attach your post as a word document unless instructed to do so or you will not receive credit. Type directly in the text box unless otherwise instructed. You may find it helpful to cut and paste into the text box from a word document. Posts should be professional, courteous, and use correct grammar and spelling. You may link anything you wish or attach articles relevant to the discussion, but this is not required unless specifically stated in the direction for that specific discussion. Include in-text citations and reference list (in APA

7th Edition format) when you reference materials from within or from outside course materials. This means that most initial posts will have in-text citations and references. Please remember to use 'netiquette' in your posts and be respectful of your peers.

Bloom's Taxonomy

Lower order thinking skills			Higher order thinking skills		
remember	understand	apply	analyze	evaluate	create
<u>Key words</u> Define, label, name, recall, duplicate, spell, list, select, recognize, identify, retrieve <i>Ex: Can you recall...?</i>	<u>Key words</u> Interpret, clarify, paraphrase, exemplify, illustrate, classify, categorize <i>Ex: How would you rephrase the meaning...?</i>	<u>Key words</u> Demonstrate, apply, illustrate, utilize, model, build <i>Ex: What ideas apply..?</i>	<u>Key words</u> Categorize, dissect, discover, inspect, contrast, conclusion, examine, survey <i>Ex: What conclusions can you draw...?</i>	<u>Key words</u> Validate, opinion, support, judge, decide, compare, appraise, value <i>Ex: Are there any errors...?</i>	<u>Key words</u> Compose, develop, hypothesize, invent, organize, produce, propose, design <i>Ex: How would you solve the following...?</i>

Discussion Board Grading Rubric

Initial Posting Rubric: Ideas and Organization	
* Proficient – well-developed; answers the questions asked; introduces new ideas; incorporates readings, meets the word count requirement, & initial post is on-time	4-5
* Basic – primary ideas are posted; organization of ideas is weak, vague, or largely conjecture; post is on-time or late, but within the week of the assigned discussion	3-3.9
* Minimal/Below average – poorly developed: does not add to the discussion; excludes answers to the essential questions posted; late work	1-2.9
* Unacceptable – failure to submit	< 1
TOTAL:	5

Response Posting Rubric: Critical Thinking and Integration	
* Proficient– responses to colleagues are articulate; specific references made to course content that indicate an understanding and synthesis of key principles under study; reflects critical thinking; meets the word count requirement & post is on-time	4 – 5
* Basic – responses to colleagues as required, but organization of ideas or support is weak, vague, or largely conjecture; post is on-time or late, but within the week of the assigned discussion	3-3.9
* Minimal/Below average – responses to colleagues, but support is lacking, and response is based on conjecture with no reference made to readings or coursework; late work	1-2.9

* Unacceptable – failure to submit	< 1
TOTAL:	5

10 points total per discussion

If you are not familiar with Bloom's Taxonomy, here are a few sites you can visit:

<https://cft.vanderbilt.edu/guides-sub-pages/blooms-taxonomy/>

<http://www.nwlink.com/~donclark/hrd/bloom.html>

<http://www.teachthought.com/critical-thinking/blooms-taxonomy/249-blooms-taxonomy-verbs-for-critical-thinking/>

APPENDIX B

Instrument Evaluation Presentation

Overview and Getting Started: This is a doctoral level scholarly presentation on a selected instrument that can be completed individually or in a group of no more than 4 students. If working in group, one member of the group is responsible for submitting the presentation for a grade in the submission area and posting it in the discussion board. The team will receive a group grade for the presentation and for having the presentation submitted and posted in the correct format and on time (see course calendar). Use the criteria below to create a 20-25 minute presentation.

Researching instruments can be time consuming and difficult, often requiring a deep dive into older literature. Please plan ahead do not delay getting started. All the materials related to the presentation must be posted on time by the due date to ensure all materials load and play correctly because the class discussion that week depends on student presentations. Planning ahead is critical and it is the students' responsibility to ensure materials are posted and accessible on time. Late work is not acceptable and can result in a zero.

Purpose: The purpose of this presentation is to explore and evaluate a selected instrument or measurement tool that measures a specific concept relevant to nursing science. It may be helpful to select one that measures a concept related to topics being considered for your DNP project but this is not required.

Form and Process: Before beginning the development of the presentation, complete the sign up page with the selected concept and instrument with your group information for faculty approval. Many of the items in the grading rubric are covered in the Module 2 materials with posted resources demonstrating instrument evaluation and development. Additional content related to the rubric is covered throughout the semester in the Intellectus materials. Refer back to those materials as needed.

This presentation is to be formatted as a voiced PowerPoint presentation of no more than 20-25 minutes with the camera on. If you are working in a group is not required for all group members to be on camera but you may opt for this. References are to be included either within the slide if appropriate and at the end of the PowerPoint presentation as unvoiced slides. **Presentations must be posted in the assigned discussion board and submitted in the designated submission area using Canvas Studio. If you do not use this format, your presentation will not be graded and will not be considered as submitted or posted. Directions for using Canvas studio are posted at the end of the syllabus and please remember to disable the comments.**

Resources: Contact the course faculty, the library, CTLE or the helpdesk early if you need assistance. Library resources including how to schedule a consultation with a health science librarian are posted in Canvas and CTLE and helpdesk contacts are posted in this syllabus.

In researching the concept, it may be helpful to look in the literature for any concept models, concept analyses, or clarifications on the concept. This type of literature often cites theoretical concept definitions and important essential elements or domains of the concept. In addition to researching how the selected instrument has been used and tested, research into your instrument may require going back many years and looking into the history of the development and testing of the instrument. You need to review the items of the instrument in order to evaluate it and give examples and to relate important conceptual domains to the subscales of the instrument. In addition to databases and journal articles, this content may appear in books or websites (some instruments may have their own websites). Your research in the literature on the instrument can assist you in identifying the strengths and weaknesses of the instrument including the results of instrument testing including validity and reliability.

Below is a list of potential examples to use for this presentation but you are not limited to these. Please reach out to faculty if you have other interests to explore.

*The concept of interest is **pain** in pediatric patients, and the selected instrument is the **Wong-Baker Faces Pain Rating Scale***

*The concept of interest is **patient satisfaction** as an indicator of quality of care, the selected instrument is **Press Ganey Score**.*

*The concept of interest is **depression** among adults, and the selected instrument is **Beck's Depression Inventory***

*The concept of interest is the **level of consciousness** in acute trauma patients, the selected instrument is **Glasgow Coma Scale***

*The concept of interest is **burnout** among nurses, and the selected instrument is the **Maslach Burnout Inventory**.*

*The concept of interest is **pain** in patients unable to self-report, and the instrument is the **Behavioral Pain Scale Non-Intubated Instrument (DNS-NI)**.*

*The concept of interest is **job satisfaction** in nursing and the instrument selected is the **McClosky/Mueller Satisfaction Scale (MMSS)**.*

*The concept of interest is **adherence** to medication regimens and the instrument selected is the **Morisky Medication Adherence Scale (MMAS-8)**.*

*The concept of interest is **anxiety** in adults and the instrument selected is the **Generalized Anxiety Disorder Scale (GAD-7)***

*The concept of interest is assessing **addiction** in adults and the instrument selected is the **Addiction Severity Index (ASI)***

Grading Criteria	Points
<i>Introduction</i> Provide a brief introduction and overview of the concept of interest and the selected instrument as it relates to nursing science and why it is important. Include: <u>A brief introduction to the selected concept.</u> This concept is what the selected instrument measures. <u>Significance of the Concept to Nursing Discuss</u> the significance of the concept or construct within the context of nursing. Defend why is it important to explore in nursing science (e.g., healthcare costs, patient suffering, consequences). Include any support or statements from relevant authoritative professional organizations or epidemiological stats or costs. <u>Clearly Identify the Selected Instrument that Measures the Concept of Interest</u> Clearly identify the specific instrument you have selected to explore and evaluate as the basis of this presentation that represents the identified concept of interest. <i>“This presentation explores and evaluates X instrument as a measure of x concept in the context of x area of interest...” This presentation explores and evaluates the Wong-Baker Faces Pain Rating Scale as a measure of the complex concept of pain in pediatric patients...</i>	15
<i>Concept Overview</i> <u>Conceptual Development Discuss</u> how well established the concept is in the literature (e.g., relevant history or duration in literature, the evolution of the concept, how well studied and analyzed) and how the concept has been explored in the literature (e.g., specific populations, diagnoses, settings or other contexts such as a specific theory association). <i>Example: Developed by Spitzer, Kroenke, Williams, and Löwe in 2006, the GAD-7 was created by analyzing 13 items from a large primary care sample to select the 7 items most strongly correlated with GAD severity. The GAD has been extensively tested in literature as a 7-item, self-report tool for screening and severity tracking of generalized anxiety disorder, particularly in primary care settings. Its history and application has expanded from initial validation in primary care to use in various populations, languages, and settings, including online platforms...</i>	25

Briefly identify what (if any) main theories or models/frameworks are associated with your construct or concept in general or in relation to your area of interest.

Example: The theoretical basis of the Beck Depression Inventory (BDI) is Beck's Cognitive Theory of Depression, developed by psychiatrist Aaron T. Beck in the 1960s. It marked a significant shift in psychology by proposing that depression is rooted in a person's distorted thought patterns rather than the unconscious conflicts suggested by psychoanalytic theory.

Concept Definitions and Essential Elements **Be specific** in listing how the concept is defined in the literature. Discuss if and how the concept is **theoretically defined** in the literature. Discuss any other definitions such as those supported by authoritative organizations, leading research, or operational definitions. It is permissible and advisable to use quotes if needed to discuss definition(s). Most instrument developers will identify specific theoretical or other definitions of the concept of measure.

In reviewing the concept definition(s), identify specific important essential elements or attributes of the concept and its associated domains, and/ or aspects of the concept that are important and that an appropriate instrument should capture.

For example, if you were looking at instruments measuring depression, you would want to see if there were any items on the instrument that addressed hopelessness or sadness as these are domains of the larger concept of depression. If you were looking at an instrument measuring the level of consciousness, it would be important to include motor function items and other domains. Often these are in the subscales of instruments and in how the concept is defined. If there are multiple definitions, identify the most appropriate definition of your concept based on the context of your selected instrument and defend your choice. This is often the definition identified by the instrument developers.

Relationships Associated with the Concept Note any important relationships your concept has with other concepts or variables in the literature that would be helpful in measurement. These relationships can be associations, antecedents, or effects/consequences of your concept.

For example, the concept of burnout in nursing is associated with lower levels of the concept of job satisfaction. Increasing the concept of motivation, as measured by the Motivation To Quit Scale (MTQS, an instrument that measures motivation) is associated with positive health behaviors such as smoking cessation (effect). Another example of a relationship is that the concept of anxiety is a component of PTSD (a larger concept within which anxiety is a component).

Distinction From Similar Concepts If applicable, identify and briefly discuss any similar concepts that are important to distinguish from your concept of interest in an instrument and what essential elements of the concept and/or notable relationships of your concept can help with this distinction. *For example, measuring patient satisfaction while being a component of overall quality of care is not the same as measuring the quality of care and should not be conflated with the larger concept quality...Depression is not the same concept as anxiety because of differing theoretic definitions and essential elements (list and discuss)- though there is some overlap in both of these examples (and get into the specific overlaps of conceptual space). Adherence is similar to compliance but is not the same and should not be used*

interchangibly. Adherence is an active, collaborative choice where a patient follows a mutually agreed-upon treatment plan, whereas compliance is a passive, one-way behavior of simply "doing as told". Adherence emphasizes patient empowerment and partnership, while compliance implies authority...

Instrument Appraisal (the main part of presentation)

30

Identify the specific instrument you have selected to explore, evaluate, and defend as a measure of the selected concept. Include:

Instrument Description Provide a general description of the instrument. Include the purpose or intended use of the instrument, any important historical background, the settings or populations of use, and any other pertinent general information such as if special training or permissions are required for use. Describe the number and type of instrument items (e.g., Likert scale, rating system, and/or psychometric, clinical, etc.), the length of time to administer or complete the instrument, any subscales or domains of the instrument.

Discuss how well-established the instrument is in the literature and/or clinical use. *For example, if the instrument has been adapted to use with different populations or if there have been changes or revisions to the instrument items over time.* Identify if the instrument been used extensively in the literature and tested or if is it still undergoing development. Is this a newer instrument still being developed or an older well-established instrument?

Provide a brief general summary of how or if the instrument has been tested or used in the literature as related to nursing.

Identify any theories used in testing or development the selected instrument including what the developer(s), or perhaps other researchers, of the instrument identify as an associated theory for the basis of the instrument is it atheoretical.

Instrument Appraisal and Critical Evaluation Appraise how well the instrument captures the concept it claims to measure through a discussion how well the instrument fits and aligns with the associated theory and definitions of the concept. This section depends on the above section on the concept that the instrument measures and associated theory being clear and developed. It may be helpful to go back and review the literature on the theory, theoretic definitions in concept models or analyses you outlined above and also reviewing the instrument development, domains/subscales and items of the instrument and in order to make connections between the instrument and the definition(s) that demonstrate consistency and alignment.

Identify the theoretic definition of the concept the instrument measures as identified by the developer(s) or others -this may require a deep investigation into the literature. If no theory or definition is offered by the developers, this is important to note and you will still need to make comparisons of the instrument to established theory and established definitions of the concept in the literature. Specifically comment on how a conceptual and/or theoretic basis is or is not well cited by the instrument developers or in research on the instrument. If there is no identified theory or appropriate definition identified in research using the instrument or developing it, this may cause problems with validity or generalizability and this would be a notable weakness of your

instrument in the conclusion section below.

Evaluate what is identified in the paragraph above as the theoretic basis and definition of the concept associated with the instrument. Evaluate and discuss if the instrument is consistent with the theory and what is presented in the literature and the definition of the concept. This will require visiting the theory and definitions and making comparisons to the instrument. Support your assessment with specific information on the concept and instrument with specific information about the concept and instrument items. Give examples, identify and comment on some specific instrument items and domains or subscales on the instrument that align well (or not) with the definition of the concept being measured, the identified essential elements and/or domains of the concept, or other aspects of the concept. *For example, it would be good to note that identified items capture essential elements of the conceptual definition.* Recall that conceptual domains or elements will often correspond with instrument subscales/domains and/or items so this is a point of comparison. If the instrument has subscales or domains with that represent different domains, elements, attributes, or aspects of your concept of measure, comment and give examples on the congruency (or lack of) of any subscale or domain items on the instrument with the concept and concept domains. To discuss these connections, it may be helpful to again review the instrument, the literature about the development, evolution, or use of the instrument, and/or review specific items of the instrument to determine fit with the definition(s).

Instrument Measurement Properties Discuss the specific measurement properties of the instrument and focus on the types of reliability and validity (see module 2 and other Intellectus course materials) covered in this course that are applicable to your instrument. Discuss and evaluate how well are reliability and validity are established. Be specific about the type of reliability and validity when appropriate. What is the level of statistical evidence (*For example, what specific measures of reliability are reported such as Cronbach's alpha or KR20, internal consistency, etc.?* What specific measures of the degree of validity are reported?), and non-statistical evidence (such as face validity and you can comment to your own idea of the face validity if none is reported, and/or your subjective judgment about the degree of content validity of the instrument) to support this and why? Sometimes instruments may be more valid or reliable in specific settings or populations and this would be important to note.

What specific subscales of the instrument have been tested? Remember that newer instruments may not have high degrees of validity if instrument development is ongoing, you must first establish reliability in order to establish validity. You do not have to address every single type of validity and reliability, just what is available and/or appropriate.

Instrument Fit and Comparison Briefly discuss, compare, and contrast your instrument to other instruments if others are available that measure the same concept. Briefly identify and discuss what other instruments (if any) are available in the literature to measure your concept of interest and how these have been used and how your selected instrument compares. If there are numerous instruments and too many to briefly discuss, just state that there are numerous instruments and select at least 1-2 instruments to discuss and compare that may be possible fits for your concept and how your selected instrument compares as a fit to measure the concept.

Remember there is no perfect instrument- valid endorsements may be based on a

numerous mix of reasons such as the congruency of the instrument with the theoretic definition of the concept, established validity and reliability, use of the instrument with a particular population or disease, ease of use (shorter is less time-consuming, special training can be a problem, etc.,) and remember each instrument is in various stages of development so the best pick may not always be the most valid/reliable if further use is working toward instrument development however, it is reasonable to support the use of an instrument because it is the most tested, reliable, and valid and it is also reasonable to support the use of an instrument that may not be as tested because it is a better fit for your concept of focus based on congruency with how the concept is defined or alignment with a more specific context.	
<i>Conclusion</i> State your overall impression of the quality instrument based on the evidence you presented. Summarize the major strengths and weaknesses of the instrument. <i>For example, established reliability and validity measures, a shorter length or easy to use instrument could be a strength. Weakness examples may include having to get permission or pay for use of the instrument, the instrument hasn't been tested in some relevant populations, training is required for use...etc.</i> Give any specific suggestions for your selected instrument for further development. <i>Some potential items to address here could be: Does it need further testing to better establish validity and/or reliability? Would using the instrument in other areas/settings/populations contribute to the development of the instrument? Does it need to be tested more in order to see if findings are reproducible in other populations or settings that are relevant to the concept it measures to contribute to reliability and validity? Does the concept that the instrument measures need further clarification or a theoretic definition to increase generalizability.</i> Remember no new information should appear to the viewer in the conclusion. Use this section to summarize important points of the findings and make recommendations based on the findings.	15
<i>Doctoral Level Presentation</i> The presentation adhered to the time limit while demonstrating organization, clarity, and logical flow of ideas and slide structure, clarity, grammar, content, spelling, APA format, correct statistical and research terminology, and professionalism. A couple of reminders: use limited quotes per APA and quotes should only be used when discussing definitions or when they add something to the presentation; avoid casual language or trite expressions like "we as nurses in the medical field", and avoid using "prove" or "proof" and use other wording such as "supports", "established", "challenges" ...	15
<u>Total</u>	100

*Instrument Evaluation Assignment may count as up to 10 DNP practice hours.

How to create a voice over Power Point & Post with Canvas Studio

- **Please know that the following instructions are simply one way to create a presentation in Canvas Studio format. There are other ways/uses for Canvas Studio that can be found at the following link:** <https://community.instructure.com/en/kb/canvas-studio-guide>

First: make slides in PowerPoint the way you want them to appear.

When ready to record your voice, go to “slideshow,” “record slide show” on the top navigation bar. Record your narration over each slide. Make sure your microphone and camera are on.

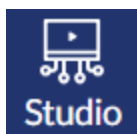
Save the presentation.

Second: save the presentation *again* with a different name and in the form of an Mp4. This is an option when you choose “save as” and then choose the **Mp4** option that appears in the drop-down box. This step can take several minutes or more depending on how long the recording is. Be sure to leave yourself enough time for this part of the process.

How to Upload Your Presentation to Canvas Studio:

Now that your presentation PPT slides and voice over are ready and in mp4 format, you can upload the presentation to *Canvas Studio* and then to the drop box and/or the Discussion Board.

Third: Open your Canvas dashboard where all your course shells are listed. Look for the octopus icon on the left navigation bar (see image below).



Use this link to upload your video to *Canvas Studio*. Follow the prompts.

Fourth: once your video is in *Canvas Studio*, go to the discussion board assignment and click to post your initial post, as usual. Then look for the octopus icon in the discussion board top navigation bar. Use this to open *Canvas Studio* and find your recorded presentation and post it. Be sure you click the small slide bar to turn the “Display Media Tabs” **off**. The example below shows comments turned on; you want to slide this to turn it off.

Embed options



Display media tabs



TURN THIS OFF. IT SHOULD **NOT** BE GREEN!

